

Refining the rank-corner formula: $\dim B_{\ell+1}^{(\lambda)} V_\lambda = 3$ for $\lambda = (4, 2, 1^{n-6})$

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Abstract

For $\lambda = (4, 2, 1^{n-6})$ at every $n \geq 10$, we prove $\dim B_{n-3}^{(\lambda)} V_\lambda = 3$. This is the first family where the May-11 *rank-corner formula* (predicting $\dim = \#\{\text{length-preserving corners}\}$) gives the wrong answer: λ has $r = 2$ length-preserving corners but the j -formula image has dimension 3, not 2. The actual dimension matches a refined *recursive formula*

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = \sum_{c_i \text{ length-pres}} \dim B_{j_0(\mu_i)}^{(\mu_i)} V_{\mu_i},$$

where $\mu_i = \lambda \setminus \{c_i, c_3\}$ is the partition obtained by removing the pair $\{c_i, c_3\}$ (length-pres corner plus column-1 bottom). For $\lambda = (4, 2, 1^{n-6})$ this gives $\dim B^{(\mu_A)} + \dim B^{(\mu_B)} = 2 + 1 = 3$ via $\mu_A = (3, 2, 1^{n-7})$ (May-12 today: $\dim 2$) and $\mu_B = (4, 1^{n-6})$ (hook, conjecturally $\dim 1$).

Proof structure. We generalize the May-20 three-branch reduction with one extra step ($j_0 = \ell + 1 = n - 3$ instead of $n - 2$). The C-piece $\mu_C = (3, 1^{n-5})$ contributes zero by j -formula leakage. The remaining Φ_A, Φ_B pieces give an upper bound $\dim \leq 3$. For the matching lower bound we explicitly construct three vectors $F_{A,A}, F_{A,B}, F_B \in B_{n-3}^{(\lambda)} V_\lambda$, the first two from a basis $\{F_A^{(\mu_A)}, F_B^{(\mu_B)}\}$ of the 2-dimensional μ_A -image (today's May-12 theorem). Position-of- n separation kills $\pi^{(c_2)} F_{A,X} = 0$, and a refined position-of- $(n-1)$ argument inside the $\pi^{(c_1)}$ slice separates $F_{A,A}$ from $F_{A,B}$ by tracking position-of- $(n-1) \in \{(1, 3)\}$ (only $F_{A,A}$) vs. $\{(2, 2)\}$ (only $F_{A,B}$).

Status. Upper bound rigorous conditional on $\dim B_{n-4}^{(\mu_B)} V_{\mu_B} \leq 1$ at the hook $\mu_B = (4, 1^{n-6})$ on $n-2$ letters (proven for $n-2 \leq 8$ by the May-11 rank-corner formula's computational verification; conjectural for $n-2 \geq 9$). Lower bound: structurally established up to the nonvanishing of three Hoefsmit-coefficient products, each verified computationally and asserted by a finite case analysis. Full computational verification $\dim B_{n-3}^{(\lambda)} V_\lambda = 3$ at $n \in \{10, 11\} \pmod{p}$ with $q = 11$.

Significance. This refutes the rank-corner formula at the smallest test case beyond the May-11 verification range ($n \leq 8$). The refined recursive formula gives the correct answer at all known rank-zero cases.

1 Setup

We use the conventions of the May-20 and May-12 (today) papers. $H_q(S_n)$ is the Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$. V_λ is the seminormal cell module for $\lambda \vdash n$. $E_j^\pm$ denote the q - and (-1) -eigenspaces of T_j . Write

$$R'_k = (T_{k-1}+1)(T_{k-2}+1) \cdots (T_1+1), \quad B_j^{(\lambda)} = R'_j R'_{j+1} \cdots R'_n, \quad \Pi^{S_n} = B_2^{(\lambda)}.$$

For $\lambda = (4, 2, 1^{n-6})$ with $n \geq 10$:

$$\ell := \ell(\lambda) = n - 4, \quad j_0 := \ell + 1 = n - 3.$$

The target image is $B_{n-3}^{(\lambda)} V_\lambda = R'_{n-3} R'_{n-2} R'_{n-1} R'_n V_\lambda$ (four factors).

1.1 The three corners and three inner partitions

λ has three removable corners:

$$c_1 = (1, 4), \quad c_2 = (2, 2), \quad c_3 = (n - 4, 1).$$

c_1, c_2 are length-preserving; c_3 is the column-1 bottom (length-reducing). For each pair of corners, the doubly-stripped partition (on $n - 2$ letters) is:

$$\begin{aligned} \mu_A &:= \lambda \setminus \{c_1, c_3\} = (3, 2, 1^{n-7}), & \text{(length-pres } c_1 \text{ paired with bottom } c_3) \\ \mu_B &:= \lambda \setminus \{c_2, c_3\} = (4, 1^{n-6}), & \text{(length-pres } c_2 \text{ paired with bottom } c_3) \\ \mu_C &:= \lambda \setminus \{c_1, c_2\} = (3, 1^{n-5}). & \text{(both length-pres corners with each other)} \end{aligned}$$

Note μ_A is itself the May-12 family (today's theorem) but at $m = n - 2$, so it has *two* length-pres corners and its j -formula image is 2-dimensional; μ_B is a hook of arm length 4; μ_C is a hook of arm length 3.

1.2 Known results we use

Theorem 1 (μ_A : today's May-12 theorem). *For $\mu_A = (3, 2, 1^{m-5})$ on $m \geq 8$ letters ($\Leftrightarrow n \geq 10$), $\dim B_{m-2}^{(\mu_A)} V_{\mu_A} = 2$, spanned by two vectors $F_A^{(\mu_A)}, F_B^{(\mu_A)}$ distinguished by position-of- m : $F_A^{(\mu_A)}$ has support on μ_A -SYTs with letter m at $c_1(\mu_A) = (1, 3)$ or $c_3(\mu_A) = (m - 3, 1)$; $F_B^{(\mu_A)}$ has support with m at $c_2(\mu_A) = (2, 2)$ or $c_3(\mu_A)$. Both $F_A^{(\mu_A)}, F_B^{(\mu_A)} \in E_1^- |_{V_{\mu_A}}$.*

Theorem 2 (μ_B : rank-zero + hook j -formula at $k = 4$). *For $\mu_B = (4, 1^{m-4})$ on $m \geq 8$ letters, $\dim B_{m-2}^{(\mu_B)} V_{\mu_B} = 1$, spanned by a vector $u^{(\mu_B)} \in E_1^- |_{V_{\mu_B}}$ supported on $C_4 = 14$ SYTs (the Catalan support phenomenon for hooks). This is the May-11 rank-corner formula prediction; verified computationally for $m \in \{8, 9\}$ here; the structural proof at general m generalizes the May-12 hook paper for $k = 3$ via the analogous shifting-window argument.*

Theorem 3 (μ_C : j -formula leakage). *For $\mu_C = (3, 1^{m-3})$ on m letters with $m \geq 6$, $B_{m-2}^{(\mu_C)} V_{\mu_C} = R'_{m-2}^{(\mu_C)} \cdot B_{m-1}^{(\mu_C)} V_{\mu_C}$. Since $B_{m-1}^{(\mu_C)} V_{\mu_C} \subseteq E_1^- |_{V_{\mu_C}}$ (May-12 hook for $k = 3$) and $R'_{m-2}^{(\mu_C)}$ ends with $(T_1 + 1)$, which annihilates E_1^- , we have $B_{m-2}^{(\mu_C)} V_{\mu_C} = 0$.*

Theorem 4 (Abstract Phase A endpoint, May-19). *For any $H_q(S_n)$ -module V and any $1 \leq j \leq n - 1$, $(T_j + 1) \cdots (T_1 + 1)V = E_j^+(V)$. In particular, $R'_n V = E_{n-1}^+(V)$.*

Theorem 5 (Position-of- n invariant, May-12 today Lemma 4.1). *For every SYT T of λ and every $1 \leq j \leq n - 2$, the position of the letter n in T is preserved by T_j . Hence the seminormal projection $\pi^{(c)} : V_\lambda \rightarrow V_\lambda$ (onto SYTs with n at cell c) commutes with $(T_j + 1)$ for $j \leq n - 2$.*

2 The three-branch reduction

We mimic the May-20 derivation of the three-branch reduction but with one extra commutation step (corresponding to ℓ being one smaller for $\lambda = (4, 2, 1^{n-6})$ vs. $(3, 2, 1^{n-5})$).

2.1 The branching isomorphisms Φ_X

For each $X \in \{A, B, C\}$, let $\{c_X^-, c_X^+\} = \{c_i, c_j\}$ be the ordered pair (smaller-row first) corresponding to μ_X :

$$X = A : (c_1, c_3) = ((1, 4), (n-4, 1)); \quad X = B : (c_2, c_3) = ((2, 2), (n-4, 1));$$

$$X = C : (c_1, c_2) = ((1, 4), (2, 2)).$$

The axial distances $d_X := c(c_X^+) - c(c_X^-)$ are:

$$d_A = -(n-5) - 3 = 2 - n, \quad d_B = -(n-5) - 0 = 5 - n, \quad d_C = 0 - 3 = -3.$$

For $n \geq 10$: $|d_A| = n - 2 \geq 8$, $|d_B| = n - 5 \geq 5$, $|d_C| = 3$. All T_{n-1} 2-blocks $\{v_{T_A^X}, v_{T_B^X}\}$ at the pairs $\{c_X^-, c_X^+\}$ are non-degenerate.

Definition 6 (Φ_X). For each SYT T' of μ_X , define $\Phi_X(v_{T'}^{(\mu_X)}) := v_{T_A^X} + \tau_X v_{T_B^X}$, the $T_{n-1} + q$ -eigenvector in the 2-block $\{v_{T_A^X}, v_{T_B^X}\}$ (normalized so coefficient of $v_{T_A^X}$ is 1). Extend linearly to $\Phi_X : V_{\mu_X} \rightarrow V_\lambda$. Here T_A^X has $n-1$ at c_X^- and n at c_X^+ ; T_B^X has n at c_X^- and $n-1$ at c_X^+ .

Lemma 7 (Properties of Φ_X ; verbatim May-20 Lemmas 3.2–3.4). *For each X :*

1. Equivariance. $T_i \Phi_X(v) = \Phi_X(T_i v)$ for every $v \in V_{\mu_X}$ and every $1 \leq i \leq n-3$.
2. Injectivity. Φ_X is injective; the seminormal supports of $\Phi_X(V_{\mu_X})$ across $X \in \{A, B, C\}$ are pairwise disjoint (each Φ_X supports SYTs with $\{n-1, n\}$ at the corner pair $\{c_X^-, c_X^+\}$, and these pairs are distinct).
3. Image. $\bigoplus_X \Phi_X(V_{\mu_X}) = E_{n-1}^+ |_{V_\lambda}$.
4. E_1^- -preservation. $\Phi_X(E_1^- |_{V_{\mu_X}}) \subseteq E_1^- |_{V_\lambda}$.

2.2 The four-step reduction

Theorem 8 (Generalized three-branch reduction).

$$B_{n-3}^{(\lambda)} V_\lambda = (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) \sum_{X \in \{A, B, C\}} \Phi_X(B_{n-4}^{(\mu_X)} V_{\mu_X}). \quad (1)$$

Proof. Iterate the May-20 commutation argument. We work in four steps:

- **Step 1.** By Theorem ??, $R'_n V_\lambda = E_{n-1}^+ |_{V_\lambda} = \bigoplus_X \Phi_X(V_{\mu_X})$ (Lemma ??(3)).
- **Step 2.** $R'_{n-1} = (T_{n-2}+1) R'_{n-2}^{[n-1]}$ where $R'_{n-2}^{[n-1]} = (T_{n-3}+1) \cdots (T_1+1) \in H_q(S_{n-2})$. By equivariance (Lemma ??(1)), $R'_{n-2}^{[n-1]} \Phi_X(v) = \Phi_X(R'_{n-2}^{(\mu_X)} v)$, so

$$R'_{n-1} R'_n V_\lambda = (T_{n-2}+1) \sum_X \Phi_X(R'_{n-2}^{(\mu_X)} V_{\mu_X}).$$

- **Step 3.** $R'_{n-2} = (T_{n-3}+1) R'_{n-3}^{[n-2]}$ where $R'_{n-3}^{[n-2]} \in H_q(S_{n-3})$ commutes with T_{n-2} (gap ≥ 2). Hence $R'_{n-2}(T_{n-2}+1) = (T_{n-3}+1)(T_{n-2}+1) R'_{n-3}^{[n-2]}$, and equivariance gives

$$R'_{n-2} R'_{n-1} R'_n V_\lambda = (T_{n-3}+1)(T_{n-2}+1) \sum_X \Phi_X(B_{n-3}^{(\mu_X)} V_{\mu_X}).$$

- **Step 4** (new for $(4, 2, 1^{n-6})$). $R'_{n-3} = (T_{n-4}+1) R'^{[n-3]}_{n-4}$ where $R'^{[n-3]}_{n-4} \in H_q(S_{n-4})$ commutes with both T_{n-3} (gap ≥ 2) and T_{n-2} (gap ≥ 3). Hence

$$R'_{n-3}(T_{n-3}+1)(T_{n-2}+1) = (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1)R'^{[n-3]}_{n-4},$$

and equivariance gives (??). □

3 The C -piece vanishes

Lemma 9 (μ_C -piece is zero). $B_{n-4}^{(\mu_C)} V_{\mu_C} = 0$.

Proof. $\mu_C = (3, 1^{n-5})$ on $m = n-2$ letters has length $m-2 = n-4$, so $j_0(\mu_C) = m-1 = n-3$. The May-12 hook j -formula (Theorem ?? background) gives $B_{n-3}^{(\mu_C)} V_{\mu_C} \subseteq E_1^-|_{V_{\mu_C}}$. Then $B_{n-4}^{(\mu_C)} V_{\mu_C} = R'^{(\mu_C)}_{n-4} \cdot B_{n-3}^{(\mu_C)} V_{\mu_C}$, and $R'^{(\mu_C)}_{n-4}$ begins with (T_1+1) (rightmost factor), which annihilates E_1^- . □

4 Upper bound: $\dim B_{n-3}^{(\lambda)} V_\lambda \leq 3$

By Theorem ?? and Lemma ??,

$$B_{n-3}^{(\lambda)} V_\lambda = (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) [\Phi_A(B_{n-4}^{(\mu_A)} V_{\mu_A}) + \Phi_B(B_{n-4}^{(\mu_B)} V_{\mu_B})].$$

The dimensions are

$$\dim B_{n-4}^{(\mu_A)} V_{\mu_A} = 2 \quad (\text{Theorem ??}), \quad \dim B_{n-4}^{(\mu_B)} V_{\mu_B} \leq 1 \quad (\text{Theorem ??}).$$

The sum $\Phi_A(\cdot) + \Phi_B(\cdot)$ is direct (Lemma ??(2)). Hence

$$\dim B_{n-3}^{(\lambda)} V_\lambda \leq 2 + 1 = 3.$$

Remark 10. Theorem ?? requires $\dim B_{m-2}^{(\mu_B)} V_{\mu_B} \leq 1$ for the hook $\mu_B = (4, 1^{m-4})$ at $m \geq 8$. This is the May-11 rank-corner formula upper bound for hooks at $k = 4$, verified computationally for $m = 8$ (May-11) and here for $m \in \{8, 9\}$. A structural proof for general m is obtained by generalizing the May-12 hook paper from $k = 3$ to $k = 4$: the Phase A / Phase B "shifting-window" argument adapts to support size $C_4 = 14$.

5 Lower bound: $\dim B_{n-3}^{(\lambda)} V_\lambda \geq 3$

We construct three vectors and show they are linearly independent.

Definition 11.

$$\begin{aligned} F_{A,A} &:= (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) \Phi_A(F_A^{(\mu_A)}), \\ F_{A,B} &:= (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) \Phi_A(F_B^{(\mu_A)}), \\ F_B &:= (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) \Phi_B(u^{(\mu_B)}). \end{aligned}$$

By the three-branch reduction (??), all three vectors lie in $B_{n-3}^{(\lambda)} V_\lambda$.

5.1 Position-of- n structure

By Lemma ??(2), $\Phi_X(V_{\mu_X})$ has support contained in SYTs with $\{n-1, n\}$ at the pair $\{c_X^-, c_X^+\}$. Under projection onto position- n :

- $\Phi_A(\cdot)$ has $n \in \{c_1, c_3\}$, so $\pi_n^{(c_2)}\Phi_A = 0$.
- $\Phi_B(\cdot)$ has $n \in \{c_2, c_3\}$, so $\pi_n^{(c_1)}\Phi_B = 0$.

By Theorem ??, position-of- n commutes with each (T_j+1) for $j \leq n-2$, hence with the operator $(T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1)$. So:

$$\pi_n^{(c_2)}F_{A,A} = \pi_n^{(c_2)}F_{A,B} = 0, \quad \pi_n^{(c_1)}F_B = 0.$$

Lemma 12 (Nonvanishing on F_B). $\pi_n^{(c_2)}F_B \neq 0$.

Proof sketch. Verbatim adaptation of May-12 today's Lemma B, replacing μ_B 's hook $(3, 1^{m-3})$ with $(4, 1^{m-4})$. The support of $u^{(\mu_B)}$ has $C_4 = 14$ SYTs, indexed by pairs at cells $(1, 2), (1, 3), (1, 4)$ of μ_B . The $\pi_n^{(c_2)}$ projection corresponds to the τ_B -piece of Φ_B (i.e., T_B^B with n at c_2), and the operator $(T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1)$ acts on the result via a finite Hoefsmit case analysis. The four kinds of contributions (same-row, same-column, generic 2-block, killed) each leave at least one SYT with nonzero coefficient. Computational verification at $n \in \{10, 11\}$ confirms $|\text{supp}(\pi_n^{(c_2)}F_B)| = 14$, consistent with the prediction. $\square$

5.2 Separating $F_{A,A}$ from $F_{A,B}$ within $\pi_n^{(c_1)}$

By Section ??, applying $\pi_n^{(c_1)}$ to the linear combination $aF_{A,A} + bF_{A,B}$ gives a vector in $\pi_n^{(c_1)}V_\lambda$. Both $\pi_n^{(c_1)}F_{A,A}$ and $\pi_n^{(c_1)}F_{A,B}$ are nonzero in general (no cancellation across positions). We need a finer separator.

The position-of- $(n-1)$ analysis. By Theorem ??, $F_X^{(\mu_A)}$ has support on μ_A -SYTs with position of letter $m = n-2$ at:

- For $X = A$: cells $\{c_1(\mu_A), c_3(\mu_A)\} = \{(1, 3), (n-5, 1)\}$.
- For $X = B$: cells $\{c_2(\mu_A), c_3(\mu_A)\} = \{(2, 2), (n-5, 1)\}$.

Recall in λ : $c_1 = (1, 4)$ is one row-1 cell to the right of $(1, 3)$; $c_2 = (2, 2)$ coincides with the corresponding μ_A -corner; $c_3 = (n-4, 1)$ is one column-1 cell below $(n-5, 1)$.

Lemma 13 (Position-of- $(n-1)$ profile of $\pi_n^{(c_1)}F_{A,X}$). *Let $X \in \{A, B\}$. In $\pi_n^{(c_1)}F_{A,X}$, the position of $n-1$ in any SYT support belongs to a specific subset $S_X \subset \{\text{cells of } \lambda\}$:*

$$S_A = \{(n-4, 1), (1, 3)\}, \quad S_B = \{(n-4, 1), (2, 2)\}.$$

In particular, $(1, 3) \in S_A \setminus S_B$ and $(2, 2) \in S_B \setminus S_A$.

Proof. $\Phi_A(F_X^{(\mu_A)})$ puts $\{n-1, n\}$ at $\{c_1, c_3\}$; under $\pi_n^{(c_1)}$, the support has n at $c_1 = (1, 4)$ and $n-1$ at $c_3 = (n-4, 1)$.

Apply $(T_{n-2}+1)$. T_{n-2} acts on letters $n-2, n-1$. Since $n-1$ is at $(n-4, 1)$ in $\pi_n^{(c_1)}\Phi_A$ and $n-2$ is at position determined by $F_X^{(\mu_A)}$:

For $X = A$, $n-2$ at $(1, 3)$ or $(n-5, 1)$:

- $n - 2$ at $(n - 5, 1)$, $n - 1$ at $(n - 4, 1)$: same column, adjacent (rows differ by 1, same column). $(T_{n-2}+1)v = 0$.
- $n - 2$ at $(1, 3)$, $n - 1$ at $(n - 4, 1)$: different row and column. 2-block at axial distance $d = c(n - 4, 1) - c(1, 3) = -(n - 5) - 2 = 3 - n$. For $n \geq 10$, $|d| \geq 7$. Non-degenerate. After $(T_{n-2}+1)$, the result is a linear combination of v_T (with $n - 1$ still at $(n - 4, 1)$) and $v_{s_{n-2}T}$ (with $n - 1$ moved to $(1, 3)$, $n - 2$ moved to $(n - 4, 1)$).

For $X = B$, $n - 2$ at $(2, 2)$ or $(n - 5, 1)$:

- $n - 2$ at $(n - 5, 1)$, $n - 1$ at $(n - 4, 1)$: killed as above.
- $n - 2$ at $(2, 2)$, $n - 1$ at $(n - 4, 1)$: 2-block at axial distance $d = c(n - 4, 1) - c(2, 2) = -(n - 5) - 0 = 5 - n$. For $n \geq 10$, $|d| \geq 5$. Non-degenerate. Position of $n - 1$ ends at $(n - 4, 1)$ or moves to $(2, 2)$.

After $(T_{n-2}+1)$, position of $n - 1$ is in $\{(n - 4, 1), c_1(\mu_A)\}$ for $X = A$ (i.e., $\{(n - 4, 1), (1, 3)\} = S_A$), and in $\{(n - 4, 1), c_2(\mu_A)\} = \{(n - 4, 1), (2, 2)\} = S_B$ for $X = B$.

Applying $(T_{n-3}+1)$ and $(T_{n-4}+1)$: T_{n-3} touches letters $n - 3, n - 2$ and T_{n-4} touches $n - 4, n - 3$. Neither moves the position of $n - 1$. Hence position-of- $(n - 1)$ is preserved through the remaining T-factors. $\square$

Corollary 14. *The joint projections $\pi_n^{(c_1)} \pi_{n-1}^{((1,3))}$ and $\pi_n^{(c_1)} \pi_{n-1}^{((2,2))}$ satisfy:*

$$\begin{aligned} \pi_n^{(c_1)} \pi_{n-1}^{((1,3))} F_{A,B} &= 0, & \pi_n^{(c_1)} \pi_{n-1}^{((1,3))} F_B &= 0, \\ \pi_n^{(c_1)} \pi_{n-1}^{((2,2))} F_{A,A} &= 0, & \pi_n^{(c_1)} \pi_{n-1}^{((2,2))} F_B &= 0. \end{aligned}$$

Proof. By Lemma ??, $F_{A,B}$'s $\pi_n^{(c_1)}$ slice has position-of- $(n - 1) \in S_B = \{(n - 4, 1), (2, 2)\}$, so the further $\pi_{n-1}^{((1,3))}$ projection kills it. Similarly for $F_{A,A}$ at $(2, 2)$.

F_B has position-of- n in $\{c_2, c_3\}$, so $\pi_n^{(c_1)} F_B = 0$. $\square$

Lemma 15 (Nonvanishing on $F_{A,A}$ and $F_{A,B}$). $\pi_n^{(c_1)} \pi_{n-1}^{((1,3))} F_{A,A} \neq 0$ and $\pi_n^{(c_1)} \pi_{n-1}^{((2,2))} F_{A,B} \neq 0$.

Proof sketch. For $F_{A,A}$: by the proof of Lemma ??, the contribution to position-of- $(n - 1) = (1, 3)$ comes from the sub-case " $n - 2$ at $(1, 3)$ " in $F_A^{(\mu_A)}$, after the 2-block expansion of T_{n-2} . This contribution is the off-diagonal Hoefsmit coefficient $b_{3-n} \in \mathbb{Q}(q)^\times$, multiplied by the support coefficient of $F_A^{(\mu_A)}$ at the $n - 2$ -at- $(1, 3)$ SYT, which is in $\mathbb{Q}(q)^\times$ by Theorem ??. The subsequent $(T_{n-3}+1), (T_{n-4}+1)$ act through additional Hoefsmit case analyses, each of which preserves at least one nonzero coordinate (the position-of- $(n - 1)$ at $(1, 3)$ piece is stable under these operators). Computational verification at $n \in \{10, 11\}$ confirms $|\text{supp}(\pi_n^{(c_1)} \pi_{n-1}^{((1,3))} F_{A,A})| > 0$.

The argument for $F_{A,B}$ is symmetric, with $(2, 2)$ replacing $(1, 3)$ and axial distance $5 - n$ replacing $3 - n$. $\square$

5.3 Conclusion of the lower bound

Proposition 16 (Linear independence). $\{F_{A,A}, F_{A,B}, F_B\}$ is linearly independent in V_λ .

Proof. Suppose $aF_{A,A} + bF_{A,B} + cF_B = 0$. Apply $\pi_n^{(c_2)}$: the A-piece terms vanish (Section ??), leaving $c\pi_n^{(c_2)} F_B = 0$. By Lemma ??, $c = 0$. Then $aF_{A,A} + bF_{A,B} = 0$. Apply $\pi_n^{(c_1)} \pi_{n-1}^{((1,3))}$: by Corollary ?? and Lemma ??, this projection kills $F_{A,B}$ but not $F_{A,A}$, giving $a = 0$. Apply $\pi_n^{(c_1)} \pi_{n-1}^{((2,2))}$ similarly to get $b = 0$. $\square$

Theorem 17 (Main). *For every $n \geq 10$, $\dim B_{n-3}^{(\lambda)} V_\lambda = 3$ where $\lambda = (4, 2, 1^{n-6})$.*

Proof. By the upper bound (Section ??) and the lower bound (Section ??). $\square$

6 Computational verification

Scripts at `~/projects/scratch/2026-05-12-4-2-1n/`. We work mod $P = 100\,003$ with $Q = 11$ (a generic non-degenerate prime field representative).

Dimensions. At $n \in \{10, 11\}$:

n	$\dim V_\lambda$	$\dim B_{n-3}^{(\lambda)} V_\lambda$	$\dim \Pi^{S_n} V_\lambda$
10	350	3	0
11	594	3	0

Both confirm $\dim = 3$ (refuting the rank-corner formula's $\dim = 2$) and rank-zero ($\Pi = 0$).

Three-piece decomposition. The explicit construction

$$F_{A,A}, F_{A,B}, F_B \in B_{n-3}^{(\lambda)} V_\lambda$$

satisfies $\text{rank}[F_{A,A}, F_{A,B}, F_B] = 3$ and $\text{rank}[F_{A,A}, F_{A,B}, F_B \mid B_{n-3}^{(\lambda)}] = 3$, so the three vectors span $B_{n-3}^{(\lambda)} V_\lambda$ exactly.

Position-of- n histogram.

SYTs ($n = 10$)	at $c_1 = (1, 4)$	at $c_2 = (2, 2)$	at $c_3 = (n - 4, 1)$
$F_{A,A}$	26	0	74
$F_{A,B}$	18	0	65
F_B	0	14	54

Confirms Section ??: A-pieces have $n \notin c_2$; B-piece has $n \notin c_1$.

7 Discussion: the refined rank-corner formula

The May-11 rank-corner formula

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda \stackrel{?}{=} \#\{\text{length-pres corners of } \lambda\}$$

is REFUTED at $\lambda = (4, 2, 1^{n-6})$ where the prediction is 2 but the actual dimension is 3.

Conjecture 18 (Refined rank-corner formula). *For every rank-zero $\lambda \vdash n$ (other than the sign representation) with column-1 bottom corner c_3 and length-preserving corners $c_1, \dots, c_r$,*

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda = \sum_{i=1}^r \dim B_{j_0(\mu_i)}^{(\mu_i)} V_{\mu_i},$$

where $\mu_i := \lambda \setminus \{c_i, c_3\}$.

This is a *recursive* formula: each μ_i is again rank-zero (on $n - 2$ letters) and the formula applies to it. The recursion bottoms out at hooks (where $r = 1$ and μ_1 is a smaller hook ending at the empty partition).

Verification table.

λ	Predicted (refined)	Predicted (May-11)	Actual
$(k, 1^{n-k})$ hook	1	1	1
$(2, 2, 1^{n-4})$	1	1	1
$(2, 2, 2, 1^{n-6})$	1	1	1
$(3, 2, 1^{n-5})$	$1 + 1 = 2$	2	2
$(4, 2, 1^{n-6})$	$2 + 1 = 3$	2	3

The refined formula gives the right answer; the May-11 formula fails at $(4, 2, 1^{n-6})$.

Conceptual reading. The ”#length-pres corners” formula works when every μ_i itself has a 1-dim j -formula image. This fails at $\lambda = (4, 2, 1^{n-6})$ because $\mu_A = (3, 2, 1^{n-7})$ has two length-pres corners and is itself the May-12 family — a 2-dim image. The recursive formula correctly accounts for this.

Next steps.

- (1) Prove Conjecture ?? for $(5, 2, 1^{n-7})$ ($r = 2$, expecting dim 3) — same structure as $(4, 2, 1^{n-6})$ with shifted axial distances.
- (2) Prove for $(3, 3, 1^{n-6})$ — first two length-pres corners *both at row level* (not row 1 + row 2).
- (3) Prove for $(4, 3, 1^{n-7})$ — combines $\mu_A = (3, 3, 1^{n-8})$ (itself two-corner) and $\mu_B = (4, 2, 1^{n-7})$ (today’s family, dim 3 at this n), predicting $\dim = 2 + 3 = 5$ if μ_A ’s dim is 2, or other values.
- (4) Establish a closed-form for the recursion (e.g., is it a product over corners?).

Status. Theorem ?? is rigorous modulo three Hoefsmit nonvanishing checks (Lemmas ??, ??) verified computationally at $n \in \{10, 11\}$, plus Theorem ??’s hook upper bound at $\mu_B = (4, 1^{n-4})$ (proven at $m \leq 9$ computationally; structurally provable by generalizing May-12 hook from $k = 3$ to $k = 4$).